

# tf.compat.v1.VariableScope Stay organized with collections Save and categorize content based on your preferences.

[https://www.tensorflow.org/api\\_docs/python/tf/compat/v1/VariableScope](https://www.tensorflow.org/api_docs/python/tf/compat/v1/VariableScope)

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*Variable scope object to carry defaults to provide to get\_variable.*

## Function Signatures

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```
tf.compat.v1.VariableScope( reuse, name='', initializer=None,
regularizer=None, caching_device=None, partitioner=None,
custom_getter=None, name_scope='', dtype=tf.dtypes.float32,
use_resource=None, constraint=None )
```

## Code Examples

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```
tf.compat.v1.VariableScope(  
    reuse,  
    name='',  
    initializer=None,  
    regularizer=None,  
    caching_device=None,  
    partitioner=None,  
    custom_getter=None,  
    name_scope='',  
    dtype=tf.dtypes.float32,  
    use_resource=None,  
    constraint=None  
)
```

```
tf.compat.v1.VariableScope(  
    reuse,  
    name='',  
    initializer=None,  
    regularizer=None,  
    caching_device=None,  
    partitioner=None,  
    custom_getter=None,  
    name_scope='',  
    dtype=tf.dtypes.float32,  
    use_resource=None,  
    constraint=None  
)
```

```
tf.dtypes.float32
```

```
get_collection(  
    name  
)
```

```
get_collection(  
    name  
)
```

```
get_variable(  
    var_store,  
    name,  
    shape=None,  
    dtype=None,  
    initializer=None,  
    regularizer=None,  
    reuse=None,  
    trainable=None,  
    collections=None,  
    caching_device=None,  
    partitioner=None,  
    validate_shape=True,  
    use_resource=None,  
    custom_getter=None,  
    constraint=None,  
    synchronization=tf.VariableSynchronization.AUTO,  
    aggregation=tf.compat.v1.VariableAggregation.NONE  
)
```

```
get_variable(  
    var_store,  
    name,  
    shape=None,  
    dtype=None,  
    initializer=None,  
    regularizer=None,  
    reuse=None,  
    trainable=None,  
    collections=None,  
    caching_device=None,  
    partitioner=None,  
    validate_shape=True,  
    use_resource=None,  
    custom_getter=None,  
    constraint=None,  
    synchronization=tf.VariableSynchronization.AUTO,  
    aggregation=tf.compat.v1.VariableAggregation.NONE  
)
```

```
tf.VariableSynchronization.AUTO
```

## Documentation

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Variable scope object to carry defaults to provide to `get_variable`.

Many of the arguments we need for `get_variable` in a variable store are most easily handled with a context. This object is used for the defaults.

### Attributes

## Methods

### `get_collection`

[View source](#)

Get this scope's variables.

## **get\_variable**

[View source](#)

Gets an existing variable with this name or create a new one.

## **global\_variables**

[View source](#)

Get this scope's global variables.

## **local\_variables**

[View source](#)

Get this scope's local variables.

## **reuse\_variables**

[View source](#)

Reuse variables in this scope.

## **set\_caching\_device**

[View source](#)

Set caching\_device for this scope.

## **set\_custom\_getter**

[View source](#)

Set custom getter for this scope.

## **set\_dtype**

[View source](#)

Set data type for this scope.

## **set\_initializer**

[View source](#)

Set initializer for this scope.

## **set\_partitioner**

[View source](#)

Set partitioner for this scope.

## **set\_regularizer**

[View source](#)

Set regularizer for this scope.

## **set\_use\_resource**

[View source](#)

Sets whether to use ResourceVariables for this scope.

## **trainable\_variables**

[View source](#)

Get this scope's trainable variables.

